Drone Electronics, Sensors and System Architecture

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Abstract— This report describes an autonomous quadcopter as an integrated sensing, control, power and communication system. It reviews propulsion, flight electronics, navigation and perception sensors, and common interfaces. A system architecture traces power, commands and measurements, followed by an example component selection for outdoor search and rescue. Trade-offs including sensing failures, limited endurance and communication loss are discussed.

I. INTRODUCTION

A quadcopter maintains flight by varying thrust from four motor-propeller units. Its flight controller (FC) repeatedly estimates motion and commands the motors; a companion computer can interpret cameras and plan higher-level actions. The ground control station (GCS) monitors status and manages missions. A reliable design must consider the whole chain from energy supply and sensor measurements to commands and failsafes [1], [2].

Manual flight uses pilot inputs through an RC link; autonomous flight executes onboard missions or external setpoints, with the FC still controlling stability. RC mainly carries pilot commands, whereas telemetry carries vehicle state, mission data and commands between vehicle and GCS. Stabilization sensors measure vehicle motion; perception sensors measure the surrounding environment. These roles can overlap.

II. ELECTRONIC COMPONENTS AND POWER

A. Flight controller. The FC is embedded hardware containing a processor, interfaces and commonly an inertial measurement unit (IMU) and barometer. Firmware such as PX4 or ArduPilot runs on that hardware; hardware and firmware are not interchangeable concepts. The controller fuses sensor readings to estimate attitude and motion, compares them with desired values, and adjusts motor commands through feedback loops. Modes can include manual, stabilized, altitude or position control, and automatic mission modes, subject to sensor availability. Configured failsafes can trigger hold, return or landing after RC, battery or position faults [1].

B. ESCs, motors and propellers. An electronic speed controller switches battery power into timed three-phase currents for a brushless motor. The FC sends a motor demand using PWM pulses or a digital protocol such as DShot; the ESC performs electronic commutation [3]. Select an ESC with a voltage range compatible with the battery and sufficient continuous and peak current margin for the motor-propeller combination, cooling and connector rating. Four separate ESCs simplify replacement; a 4-in-1 board reduces wiring but concentrates heat and may require replacing the whole board.

Brushless motors convert electrical energy to torque; propellers convert torque to thrust. Motor KV is an approximate unloaded speed per volt, not a thrust rating. Larger propellers can increase efficiency at suitable speeds but demand more torque and clearance. Choose the motor, propeller, ESC and battery as a tested combination, ensuring total thrust exceeds loaded weight with a margin for manoeuvring and conditions.

C. Battery and distribution. Lithium-polymer batteries are common due to high power density. Cell count determines nominal pack voltage (approximately 3.7 V per cell); capacity in mAh relates to stored charge, and discharge capability and wiring constrain current. A power distribution board feeds ESCs while a power module measures voltage/current and regulators supply appropriate stable voltages to the FC, computer and sensors. Excess load, voltage sag and heat reduce usable endurance; battery monitoring and a configured low-battery action are essential.

D. GNSS and radio control. GNSS receivers estimate latitude, longitude and altitude from signals transmitted by multiple satellites; GPS is one GNSS constellation. Accuracy depends on geometry, obstruction, multipath and interference. Position hold, waypoint navigation and return-to-home depend on valid positioning and configuration. RTK uses correction data, often from a base station or network, to reduce errors; high precision depends on correction availability and a reliable solution rather than being guaranteed everywhere. RC transmitter sticks and switches map to receiver channels for pilot inputs, arming and mode selection; a lost-link failsafe must be configured and tested. ExpressLRS is one example of an RC system.

E. Companion computer. A Raspberry Pi or NVIDIA Jetson runs a general-purpose operating system and may process video, execute ROS 2 software, detect objects and plan paths. It complements rather than replaces the FC's real-time stabilization. A common serial link carries MAVLink messages between the two; the FC checks modes and conditions before acting on external setpoints [2]. The computer needs suitable regulated power, cooling and a plan for loss of communication.

III. SENSOR SYSTEMS

A. IMU, magnetometer and barometer. A gyroscope measures angular velocity and responds quickly to rotation but integration drifts. An accelerometer measures specific force, including the effect of gravity; during acceleration or vibration it is not a perfect tilt reference. Fusion combines their complementary information to estimate attitude. A magnetometer measures magnetic field to support heading estimation, but motor currents, steel and wiring can distort it; placement and calibration matter. A barometer estimates relative altitude from pressure; weather, propeller airflow and temperature can cause error. These sensors are fused with GNSS and other measurements rather than trusted in isolation [1].

B. LiDAR. A laser rangefinder estimates distance from the travel time or phase of reflected light. A 2D scanner provides ranges in a plane, while a 3D system samples multiple elevations to form a point cloud. Uses include obstacle detection, mapping, localization and height measurement. It provides direct geometry but range, field of view, weight, power and update rate constrain integration. Transparent, dark or reflective surfaces and weather can degrade returns; a single scanning plane can miss obstacles above or below it.

C. Depth cameras. A depth image assigns distance estimates to pixels; an RGB-D device combines color and depth. Stereo cameras estimate depth from disparity between two views, while other devices use time-of-flight or structured light. The RealSense D435 is a real stereo depth camera with RGB, USB and a manufacturer-listed ideal range of roughly 0.3-3 m [4]. Depth enables local mapping, object distance estimation and obstacle avoidance, but textureless surfaces, bright sunlight, occlusions and range limits can reduce reliability. A depth camera alone does not guarantee collision avoidance.

D. Optical flow. Optical flow estimates apparent image motion between frames; combined with distance to the ground and attitude information it can support horizontal velocity estimation without GNSS. It can be useful indoors, but weak light, featureless surfaces, moving scenery and high altitude degrade measurements. A downward range sensor is commonly paired with it [5].

E. Additional sensors. *Thermal infrared cameras* measure emitted long-wave radiation and can highlight people at night, but warm ground, occlusion and low resolution can cause false detections [6]. *Millimeter-wave radar* transmits radio waves and analyzes returns to estimate range and sometimes relative velocity. It can assist obstacle sensing through some adverse visibility, but has limited angular resolution, multipath and material-dependent returns. Both require appropriate mounting, power and processing.

IV. COMMUNICATION INTERFACES

Table I summarizes common interfaces. Actual protocols, electrical levels, connectors and software drivers depend on the chosen hardware; shared grounds and voltage compatibility must be verified.

TABLE I. EXAMPLE INTERFACES

Interface	Typical use	Example
UART	Serial messages	GNSS / telemetry radio
I2C	Shared low-speed bus	Magnetometer
SPI	Fast local bus	Onboard IMU
CAN	Robust shared bus	Smart GNSS / ESC
PWM	Pulse-width motor command	ESC signal input
USB	Data and peripherals	Depth camera

V. AUTONOMOUS QUADCOPTER ARCHITECTURE

Fig. 1 shows one possible arrangement. The battery feeds the ESCs through distribution and provides regulated power for electronics. The FC receives GNSS, inertial and RC inputs and sends motor demands. The companion processes sensor images and sends high-level setpoints over MAVLink; the FC returns state. An onboard telemetry radio exchanges MAVLink with the GCS. The diagram is conceptual: actual connector assignments and power rails depend on the hardware. A separate RC transmitter provides the wireless pilot link to the receiver; regulated supplies for peripherals are omitted from the drawing for clarity.

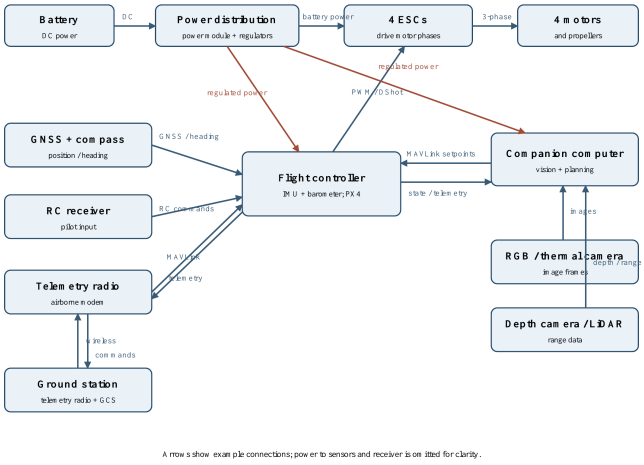


Fig. 1. Example quadcopter architecture and information flow. Original diagram.

VI. SEARCH-AND-RESCUE COMPONENT SELECTION

Flight controller: A Pixhawk-class controller running PX4 provides sensor fusion, waypoint missions and configured failsafes; verify board I/O and supported firmware [1]. **Companion:** a Jetson Orin Nano-class computer provides onboard vision inference with a power and thermal cost; its carrier and regulated supply must be sized for flight [7].

Positioning: multi-constellation GNSS plus compass enables outdoor navigation; optional RTK can improve positioning when corrections are available. GNSS may fail under canopy or near structures, so loss-of-position behavior needs testing. **Obstacle detection:** a D435 stereo depth camera provides near-field depth and RGB data [4]; its short useful range demands conservative speed and does not cover every direction. Add an appropriate longer-range sensor if required by the mission.

Human detection: RGB imagery processed on the Jetson can identify candidate people in daylight; a compatible thermal camera can support night search. The model may miss people or generate false positives; a trained human should verify detections and estimates. **Communication:** a compatible telemetry radio provides MAVLink to a GCS, and an independent RC link allows pilot intervention [8]. Confirm local frequency rules, link budget, antenna placement, range and loss behavior. Streaming video requires a separately engineered bandwidth-capable link; telemetry alone does not imply video support.

Before deployment, budget total mass, thrust, current draw, flight time, mounting vibration, environmental conditions and sensor fields of view. Validate configurations and failsafes in simulation and controlled tests, and operate under applicable local rules.

VII. CONCLUSION

A capable autonomous drone depends on a coordinated power, sensing, control and communications stack. The flight controller closes the stabilization loop while companion computing adds perception and planning. Safe mission performance depends on integration, sensor limitations, adequate power and verified responses to failure, not simply on selecting individual components.

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